

Program : Diploma in Cyber Forensics And Information security	
Course Code : 5283A	Course Title: Data and Application Security
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Understand symmetric and asymmetric cryptographic systems and cryptology through cryptography techniques and algorithms.
- Explain the hash function as an application of cryptography with the concept of message integrity and digital signature authentication.
- Understand security concepts, security professional roles, and security resources in the context of systems and security development life cycle.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge in Cryptography		Fundamentals of Cryptography and Information Security	III
Basic knowledge in Security threats		Cyber forensics and Security threats	IV

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Understand symmetric cryptographic systems and cryptology through cryptography techniques and algorithms.	14	Applying
CO2	Understand asymmetric cryptographic systems and cryptology through cryptography techniques and algorithms.	18	Applying
CO3	Explain the hash function as an application of cryptography with the concept of message integrity and digital signature authentication.	16	Applying
CO4	Describe various security threats to web applications/ servers and providing security to web servers.	10	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2				
CO2	3		2				
CO3	3		2				
CO4	3		1				

3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand symmetric cryptographic systems and cryptology through cryptography techniques and algorithms.		
M1.01	Explain basics of cryptography	2	Understanding
M1.02	Illustrate Symmetric Key Cryptographic Systems	5	Understanding
M1.03	Solve problems of Symmetric key Cryptographic Systems	7	Applying
Contents: Introduction: History of Cryptography. Mathematical background: Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES.			
CO2	Understand asymmetric cryptographic systems and cryptology through cryptography techniques and algorithms.		
M2.01	Illustrate Asymmetric Key Cryptographic Systems	4	Understanding
M2.02	Solve problems of Asymmetric key Cryptographic Systems	8	Applying
M2.03	Explain the concepts of Stream and block ciphers	6	Understanding
	Series Test – I	1	
Contents : Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs.			
CO3	Explain the hash function as an application of cryptography with the concept of message integrity and digital signature authentication.		
M3.01	Illustrate the concepts of Digital signatures	4	Understanding

M3.02	Explain various secret sharing algorithms	4	Understanding
M3.03	Illustrate random number generators	3	Understanding
M3.04	Solve problems of digital signatures, secret sharing algorithms and random number generators	5	Applying
Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description).			
CO4	Describe various security threats to web applications/ servers and providing security to web servers.		
M4.01	Explain web application security threats and defense mechanisms	4	Understanding
M4.02	Illustrate the mechanism of transmitting data	2	Understanding
M4.03	Explain vulnerabilities of access control and securing mechanisms.	4	Understanding
	Series Test – II	1	
Contents : Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls			

Text / Reference

T/R	Book Title/Author
T1	Cryptography and Network Security: Principles and Practice, Global Edition, 7/E, William Stallings, Pearson
T2	Padmanabhan T R, Shyamala C and Harini N, “Cryptography and Security”, Wiley Publications 2011.
T3	DafyddStuttard, Marcus Pinto, The Web Application Hacker’s Handbook, 2nd Edition, Wiley Publishing, Inc.

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com